DATE: 12/03/2026 DAY: __/__/____

Bilal Abid Abbasi
45961

Assignment #01

DLD (Digital Logic Design)

1) Music studio microphone system.

a)- identify which part of this process uses an analog signal and which part uses digital signal.

Analog and digital signal.

The microphone captures natural sound waves from the singer. Sound waves are continious, so this part uses an analog signal.

Digital signal:

When the sound is sent to the computer and processed, it is converted into digital from (0s and 1s) so this part uses a digital signal.

(b)- Two differences between Analog and digital signals.

DATE: __/__/____ DAY: __/__/____

	Analog signal	Digital signal
(1)	continious signal	Discrete signal (0 and 1)
(2)	Easily affected by noise	Less affected by noise.

2) Smart Security System.

a) Why Logic gates are necessary.

Logic gates are used in digital systems to process input signals and make decisions. In a security system they help decide when the alaram should turn ON or OFF base on sensor signals.

(b) Logic gate used to combine door and window sensor.

The correct gate is the OR gate.

(1)

DATE: __/__/____ DAY: __/__/____

Reason.

- if door sensor = 1 OR
- window sensor = 1

then alaram = ON

But since alarams works only when system is activated, an AND gate can also be used with the activation signal.

3) Number system conversions.

(a) Convert 345_{10} to Binary.

345 ÷ 2 = 172 R 1
172 ÷ 2 = 86 R 0
86 ÷ 2 = 43 R 0
43 ÷ 2 = 21 R 1
21 ÷ 2 = 10 R 1
10 ÷ 2 = 5 R 0
5 ÷ 2 = 2 R 1
2 ÷ 2 = 1 R 0
1 ÷ 2 = 0 R 1

Binary $(101011001)_2$

DATE: __/__/____ DAY: __/__/____

(b) Convert $(10110110)_2$ to octal.

10110110
010110110

010 = 2
110 = 6
110 = 6

Octal = (266)8 Ans.

(c) Convert $(93)_{10}$ to Hexadecimal.

93 ÷ 16 = 5 R 13
13 = D

Hexadecimal = 5D16 Ans.

(d) Convert $(11010111)_2$ to decimal.

$1 \times 2^6 = 64$
$1 \times 2^5 = 32$
$0 \times 2^4 = 0$
$1 \times 2^3 = 8$
$0 \times 2^2 = 0$
$1 \times 2^1 = 2$
$1 \times 2^0 = 1$

Total = 64+32+0+8+0+2+1
= $(107)_{10}$ Ans.

DATE: __/__/____ DAY: __/__/____

4). BCD Arithmetic

(a) 6+7

6 = 0110.

7 = 0111

```
  0110
+ 0111
  1101
```

1101 > 9 – add 0110

```
  1101
+ 0110
 10011
```

Result → 00010011(13)

(b) 8+5

8 = 1000

5 = 0101

```
  1000
+ 0101
  1101
```

1101 > 9

add 0110

```
  1101
+ 0110
 10011
```

Result → 00010011(13).

DATE: __/__/____ DAY: __/__/____

(c) 9 − 3

9 = 1001

3 = 0011

```
  1001
- 0011
------
  0110
```

Result = 6

5) Hamming code (Even parity)

Given Data

D = 1 0 1 0

Bit positions.

Position	1	2	3	4	5	6	7
Bit	P1	P2	P3	P4	P5	P6	P7

Data placement.

D1 = 1

D2 = 0

D3 = 1

D4 = 0

```
1011
0110
1001
```

DATE: __/__/____ DAY: __/__/____

Position	1	2	3	4	5	6	7
Bit	P_1	P_2	P_3	P_4	P_5	P_6	P_7

Parity calculation.

P_1 check = 1, 3, 5, 7

Bit → 1, 0, 0

odd → $P_1 = 1$

P_2 check = 2, 3, 6, 7

Bit → 1, 1, 0

Even → $P_2 = 0$

P_4 check = 4, 5, 6, 7

Bits → 0, 1, 0

odd → $P_4 = 1$

Final 7-bits Hamming code.

Position	1	2	3	4	5	6	7
Bit	1	0	1	1	0	1	0

Final code = 1011010